Home assignment: day 2 → day 5

Deadline: 2026-07-03, 16:59.

1. There are 10 standard indistinguishable dices.

   a) How many different combinations of scores are possible?
   b) How many different combinations of scores are possible if at least one dice shows 6?
   c) How many different combinations of scores are possible if no dice shows 5 nor 6?

2. There are 5 types of post-cards sold at the post-office.

   a) In how many different ways can I buy 12 post-cards?
   b) In how many different ways can I buy 12 post-cards if I would like to buy at least one post-card of every type?

3. Using Gauss summation trick or otherwise find the sum:

$$\binom{n}{0} - 2\binom{n}{1} + 3\binom{n}{2} - 4\binom{n}{3} \cdots + (-1)^n(n+1)\binom{n}{n};$$

4. Recall the mother-identity

$$(1+x)^n = 1 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n.$$

   (a) Find the first derivative of the mother-identity.
   (b) Find the value of the sum

$$\binom{n}{1} + 2\cdot 7\cdot\binom{n}{2} + 3\cdot 7^2\cdot\binom{n}{2} + \cdots + n\cdot 7^{n-1}\binom{n}{n}.$$

   (c) Find the second derivative of the mother-identity.
   (d) Find the value of the sum

$$2\cdot 1\binom{n}{2} + 3\cdot 2\cdot 9\cdot\binom{n}{3} + 4\cdot 3\cdot 9^2\cdot\binom{n}{4} + 5\cdot 4\cdot 9^3\cdot\binom{n}{5} + \cdots + n(n-1)9^{n-2}\binom{n}{n}.$$

Demo for the quiz on day 3

1. Using Gauss summation trick or otherwise find the sum

$$3\binom{n}{1} + 5\binom{n}{2} + 7\binom{n}{3} + \cdots + (2n+1)\binom{n}{n}.$$

2. Consider the equation $a+b+c=100$ with integers $a\geq 3$, $b\geq 2$, $c\geq 5$.

   How many solutions are there?